

LSGI3262 Laser Scanning – Group ■ Scanning Plan

Project title: Urban Features: 3D Point Cloud Acquisition and Semantic Segmentation of Block W Square

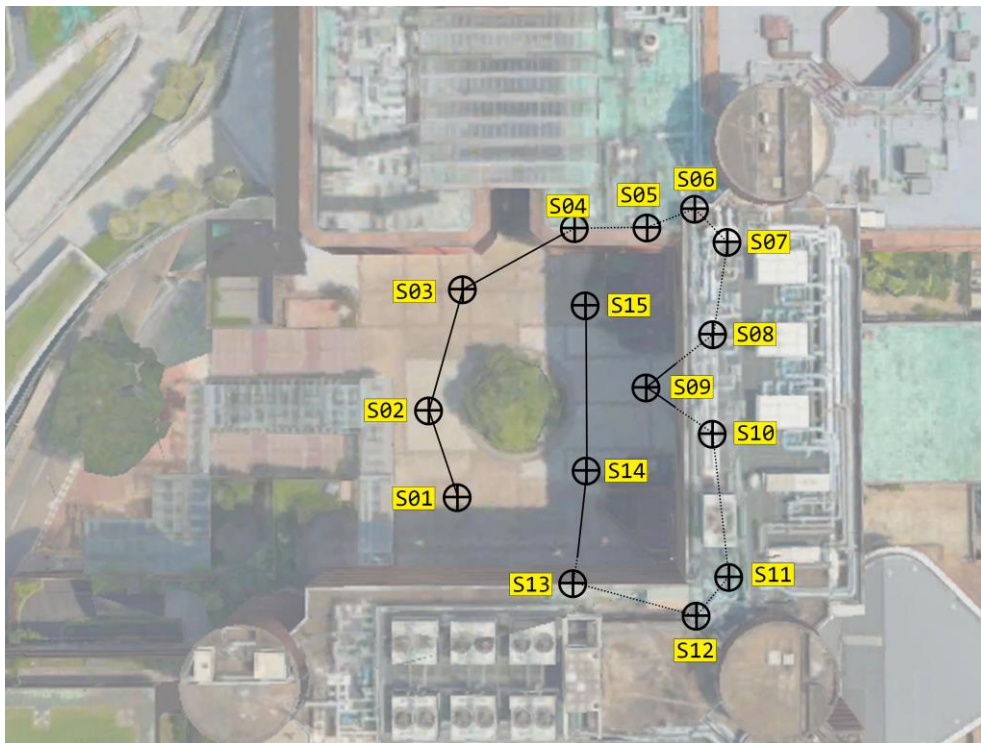
Instrument selection: TLS (Leica BLK360 G2)

Scanning location: Ho Iu Kwong and Kwok Pui Chun Square (Block W Square) and 2/F covered plaza of TU Wing (TU plaza)

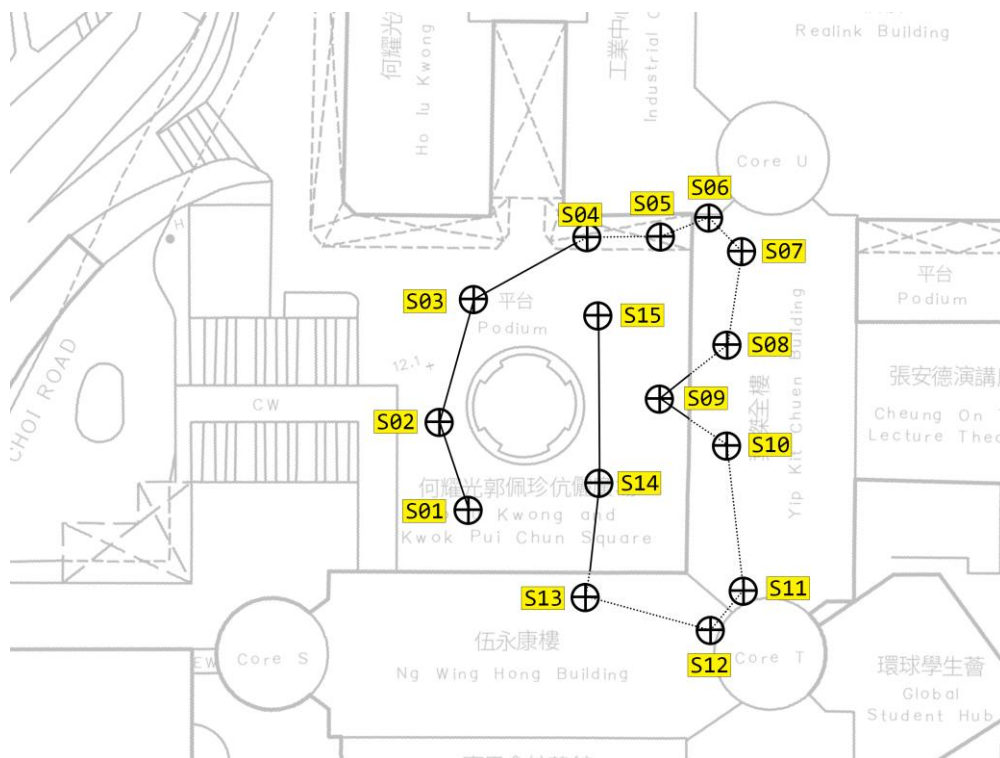
1. Introduction of group member and their roles

Member	Role	Responsibilities
██████████ ██████████d	Group leader	- Oversees overall project progress and timeline - Responsible for borrowing equipment from the Survey Store - Oversees equipment operation - Actively communicates with TA and professors - Drafted the scanning plan
██████████ ██████████d	Technical leader	- Leads all technical aspects (e.g. point cloud processing, quality control, add scans to mitigate defects) - Familiar with the Leica Cyclone software - Evaluates point cloud accuracy and quality - Creates 2D map using ArcGIS Pro - Selected the TLS scanning locations
██████████ ██████████d	Segmentation leaders & field surveyors	- Manage point cloud slicing and merging - Lead the segmentation (decides segmented classes, labeling strategy) - Segmentation workflow management - Operates laser scanner - Measures ground features using GNSS RTK for georeferencing
██████████ ██████████d		
██████████ ██████████d	Documentation leader & technical support	- Leads the project report (e.g. structure, subheadings, appendix, language) - References academic papers and literature review - Drafts introduction, rationale of survey site, and findings - Performs data preprocessing (e.g. denoising, void filling)
(All members)	(All members)	- Assist with semantic segmentation - Contribute to the project report

2. Proposed Scanning Plans



15 proposed TLS scanning locations on the Open3DHK basemap



15 proposed TLS scanning locations on the iB1000 basemap

3. Summary of the proposed scanning area

The selected area is the Block W square and the connecting covered plaza of TU Wing. The area consists of various features such as buildings, trees, planters, benches, ceilings, floors, and pillars. The point clouds can likely demonstrate the results of a TLS survey of typical urban features, while having diverse classes in the area can support semantic segmentation practice, while highlighting the challenges in semantic segmentation for urban TLS surveys.

4. Summary of the proposed scanning plan

The selected area will be surveyed using 15 separate scans, with 6 scans in the Block W Square, 3 scans in the W-TU corridor, and 6 scans in the TU plaza. Additional scans might be done ad hoc to cover the voids or increase the local point density where necessary.

5. Site reconnaissance

During the site reconnaissance, some areas of the Block W Square/ TU plaza were found not intervisible due to the pillars or other objects obstructing the view. To ensure the scanner can capture the features and merge the points, the scanning locations around the corners were close to each other. Below are the locations of the scanning locations next to the pillars.

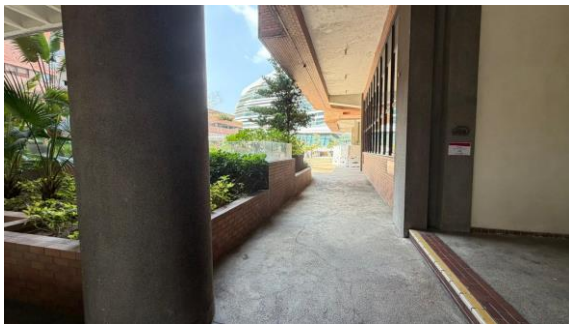


Photo from proposed S07



Photo from proposed S12

In addition to the obstruction of the pillars, the benches around the center flowerbed may not be fully visible after the previous 13 scans, so we added 2 more scans to cover these benches. Below are the locations of the proposed scanning locations.



Photo from proposed S14



Photo from proposed S15